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imbi Heidelberg
Institute of Medical
Biometry

# Computational Methods to Reduce Bias in Binary Endpoints in Dose-finding Studies

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Alexandra Balzer
Institute of Medical Biometry
University of Heidelberg

# Introduction & Previous Work

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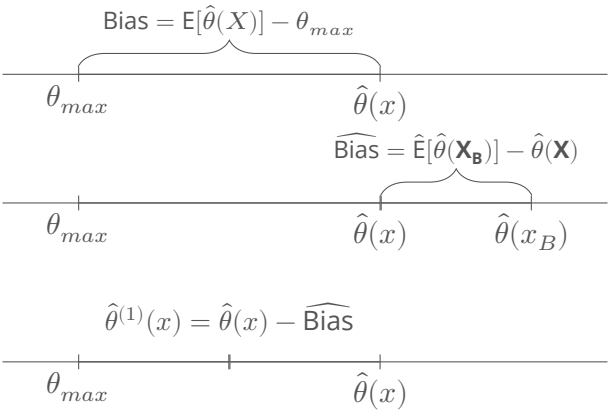
- Optimization of dose finding and selection is a key goal in oncology drug research.
- OPTIMUS Project: FDA initiative to improve dose optimization/dose finding.
- Key challenge: Accurate estimation of therapeutic effects
  - True treatment effect is often overestimated in Phase II
  - Phase III results may not confirm Phase II findings.
- Bias-correction methods were presented for normally distributed outcomes (e.g. boostrap methods) ([Zhan2024])
- No existing methods for binary endpoints identified in a literature research.

# Setup

- $I$ active treatment arms with $n$ patients.
- Response $X_{ij}$ for patient $j$ in group $i$ with $X_{ij} \sim \text{Ber}(\theta_i)$ and $i = 1, \dots, I$.
- $\mathbf{X_i} = (X_{i1}, \dots, X_{in})$ vector of responses for group $i$ and $\mathbf{X} = (\mathbf{X_1}, \dots, \mathbf{X_I})$.
- Parameter of interest is $\theta_{max} = \max(\theta_1, \dots, \theta_I)$.
- Estimator: $\hat{\theta}(\mathbf{X}) = \max[\bar{\mathbf{X}}_1, \dots, \bar{\mathbf{X}}_I]$.
- Even if $\bar{\mathbf{X}}_i = \text{mean}(X_{i1}, \dots, X_{in})$ is an unbiased estimator, $\hat{\theta}(\mathbf{X})$ is biased.

# Idea of the Bootstrap Method



# Part I - Optimization of Bootstrap Repetitions

**Aim**: Optimize the number of bootstrap repetitions $B$ for the single and double (non-)parametric bootstrap in binary endpoints, based on bias and MSE.

- Triple bootstrap is not considered in this part due to computational burden.
- The number of bootstrap repetitions $B$ is set to 80, 100, 200, 500, and 1000.

# Part II - Bayesian Hierarchical Models

**Aim**: Identify the optimal hyperprior (with corresponding parameters) for a Bayesian hierarchical model to minimize bias and MSE.

The model is specified as: $Y_i \sim \text{Bin}(n, p_i), \quad i = 1, \dots, 5$ with

$$\text{logit}(p_i) \sim \mathcal{N}(\mu, \sigma^2), \quad \mu \sim \mathcal{N}(-1.38, 100).$$

- Hyperpriors determine how much information is pooled across groups (i.e., $\sigma^2$ controls degree of shrinkage).
- The following MCMC settings are applied fo all fits: burn-in 2.000, number of chains 3, total MCMC iterations per chain: 10.000.

# Part II - Priors

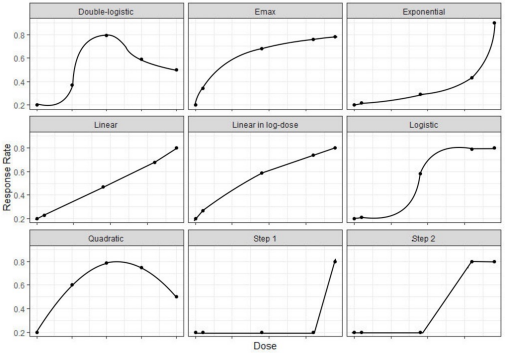
The priors and corresponding parameters to evaluate are based on [Cunanan et al.2019]:

- Unif$[0, a]$ for $\log(\sigma), \sigma, \sigma^2$ with
  - $a \in \{1, \dots, 1000\}$.
- Half-t for $\sigma$ with
  - scale parameter $s \in \{0.5, \dots, 1000\}$ and
  - degrees of freedom $df \in \{1, \dots, 500\}$.
- Gamma distribution $\Gamma(a, b)$ for $\tau = \frac{1}{\sigma^2}$ with
  - $a \in \{0.005, \dots, 2\}$, and
  - $b \in \{0.0005, \dots, 50\}$.

# General Setup

The simulation study consists of two parts (10.000 repetitions each), with the following settings:

- 9 dose response relationships; 3 or **5** doses [Ivanova et al.2012]
- Maximal response: 60% or **80%**
- Sample size: n = 20 or **40** patients per dose level



# Part I

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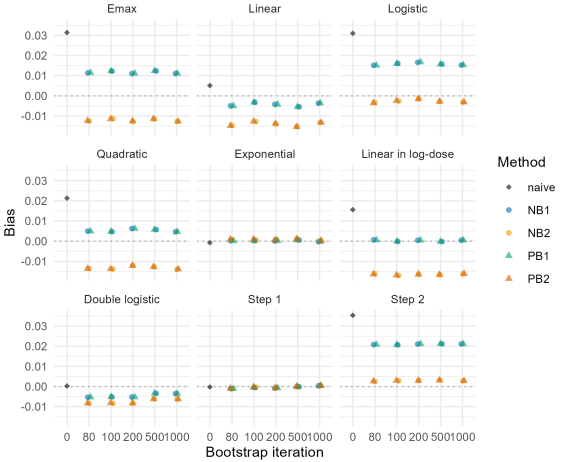
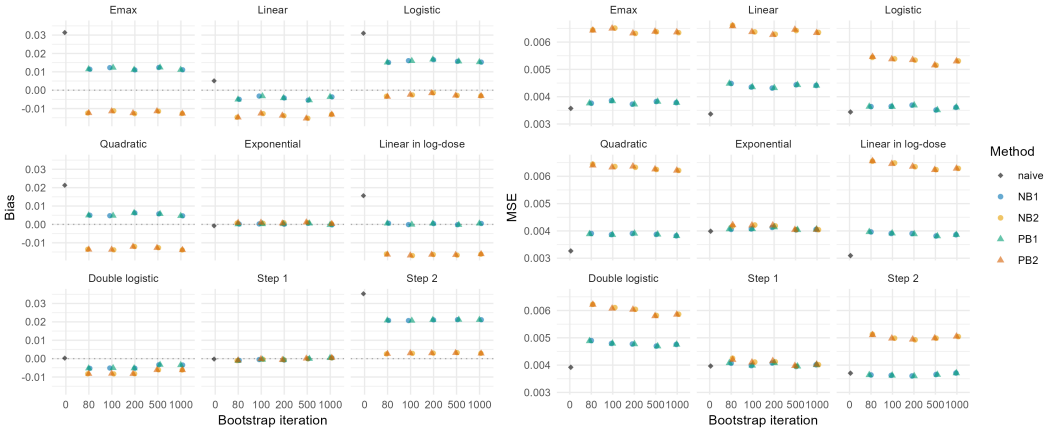

Emax
Linear
Logistic
Quadratic
Exponential
Linear in log-dose
Double logistic
Step 1
Step 2
Bias
0.03
0.02
0.01
0.00
-0.01
0 80 100 200 500 1000
Bootstrap iteration
Method
naive
NB1
NB2
PB1
PB2

# Part I


Emax
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Step 1
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MSE
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0.02
0.01
0.00
-0.01
0.006
0.005
0.004
0.003
0 80 100 200 500 1000
Bootstrap iteration
Method
naive
NB1
NB2
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# Part I: Summary

- No relevant differences in bias or MSE across bootstrap iterations.
- $B = 80$ bootstrap iterations provide adequate bias correction.
- Other simulation scenarios reached the same conclusion.

# Part II: Uniform [0, a]

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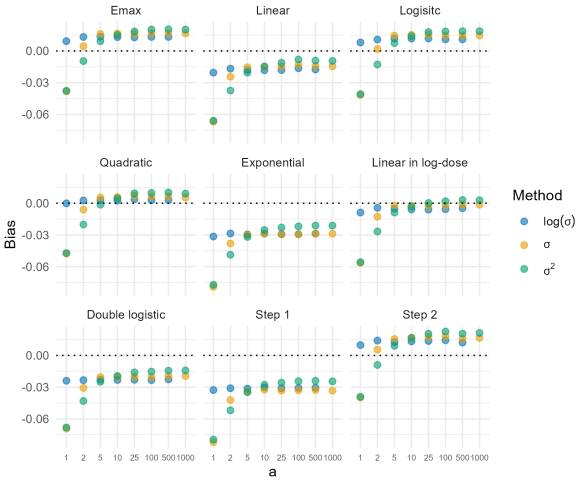

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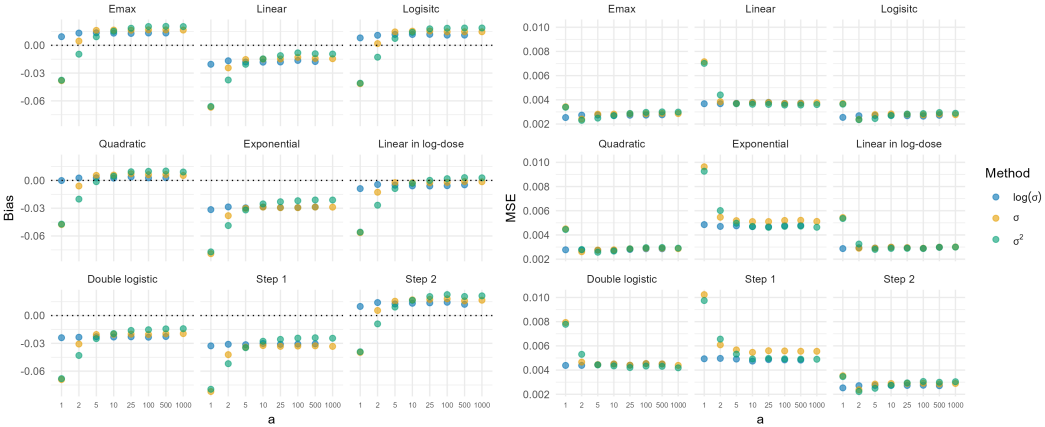

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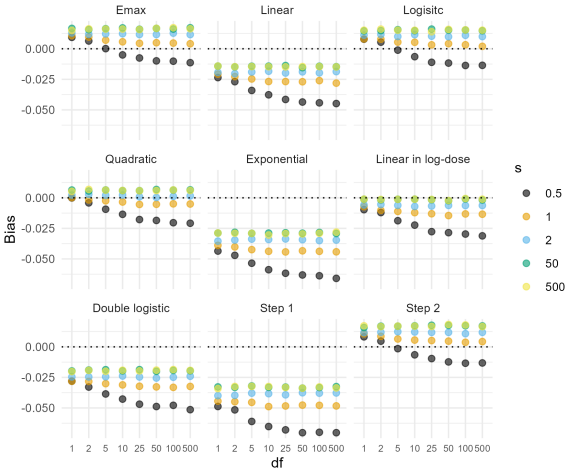

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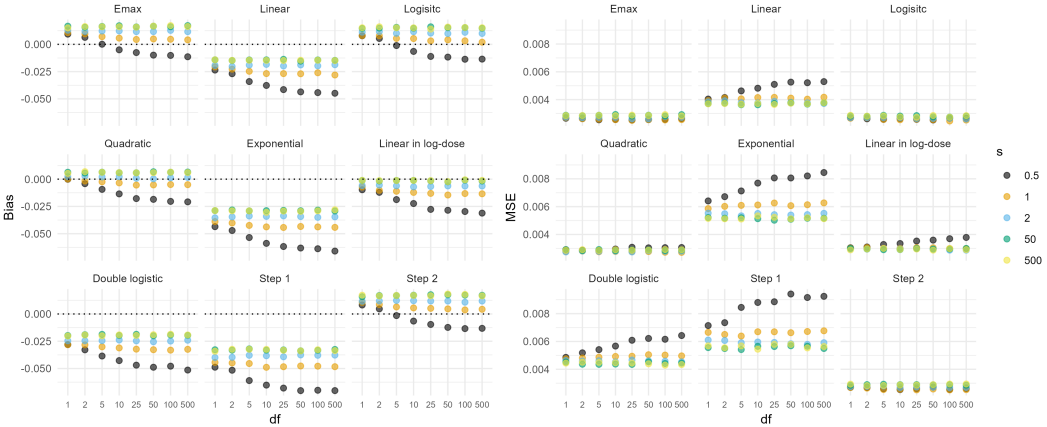
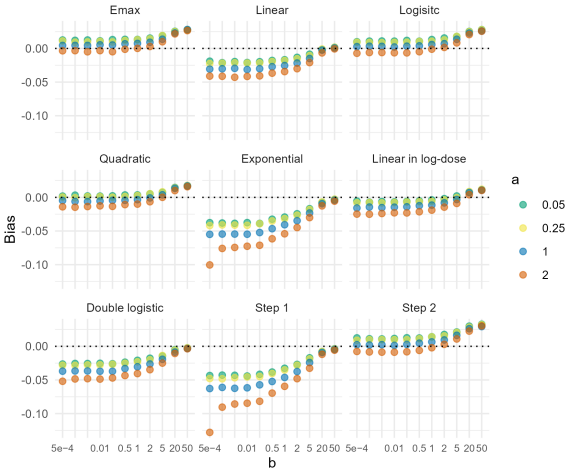


# Part II: Gamma(a,b)



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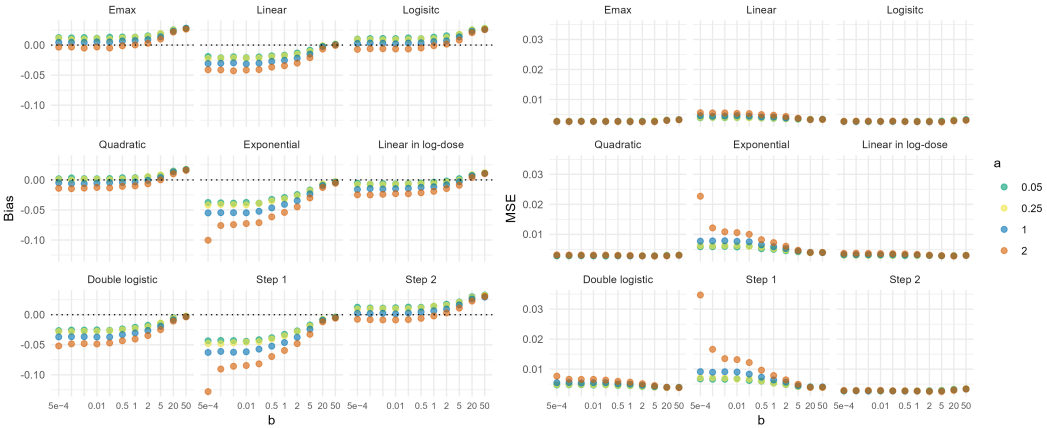

# Part II: Summary

| Method | Parameter | | | | Bias (MSE) |
|---|---|---|---|---|---|
| Uniform $\log(\sigma)$ | $a =$ | 2 | | | -0.0070 (0.00351) |
| Uniform $\sigma$ | $a =$ | 100 | | | -0.0047 (0.00371) |
| Uniform $\sigma^2$ | $a =$ | 1000 | | | 0.0003 (0.00357) |
| Half-t | $df =$ | 100 | s = | 500 | -0.0043 (0.00369) |
| Gamma | $a =$ | 0.25 | b = | 5 | 0.0004 (0.00335) |

- Uniform prior on $\sigma^2$ with $a = 1000$ and Gamma prior with $a = 0.25$ and $b = 5$ show the smallest bias.
- MSE is very similar for most parameter values and hyperpriors, except of extreme outliers.
- Prior choice depends on the true underlying dose-response scenario.

# Conclusion

- $B = 80$ bootstrap iterations are sufficient.
- Bayesian methods show promising results in reducing bias.
- Further scenarios for Bayesian methods and further methods will be investigated.

# References I

Kristen M Cunanan, Alexia Iasonos, Ronglai Shen, and Mithat Gönen. Variance prior specification for a basket trial design using bayesian hierarchical modeling. *Clinical Trials*, 16(2):142–153, 2019. doi: 10.1177/1740774518812779. URL https://doi.org/10.1177/1740774518812779. PMID: 30526008.

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